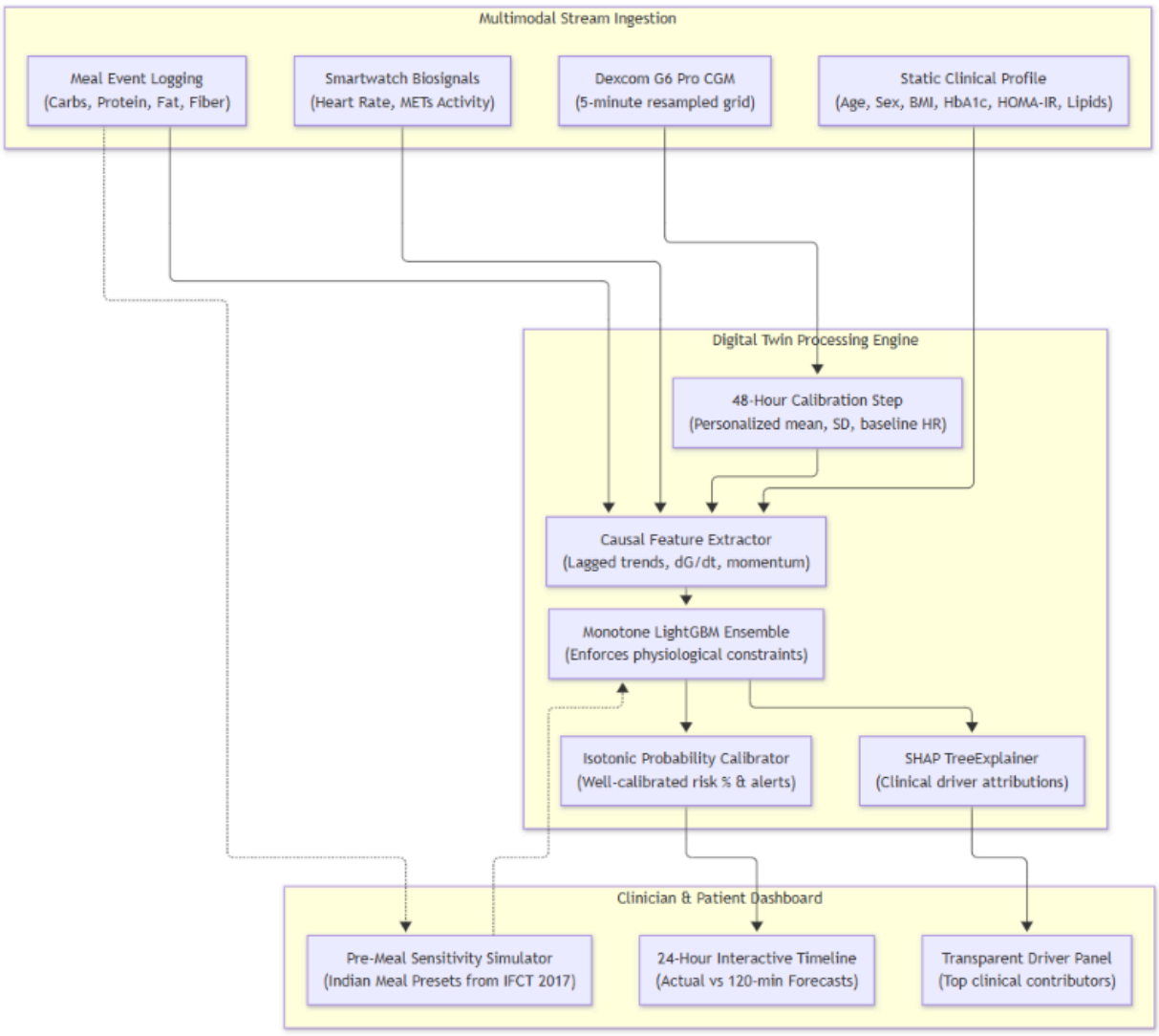

Multimodal Stream Ingestion
Meal Event Logging
(Carbs, Protein, Fat, Fiber)
Smartwatch Biosignals
(Heart Rate, METs Activity)
Dexcom G6 Pro CGM
(5-minute resampled grid)
Static Clinical Profile
(Age, Sex, BMI, HbA1c, HOMA-IR, Lipids)
Digital Twin Processing Engine
48-Hour Calibration Step
(Personalized mean, SD, baseline HR)
Causal Feature Extractor
(Lagged trends, dG/dt, momentum)
Monotone LightGBM Ensemble
(Enforces physiological constraints)
Isotonic Probability Calibrator
(Well-calibrated risk % & alerts)
SHAP TreeExplainer
(Clinical driver attributions)
Clinician & Patient Dashboard
Pre-Meal Sensitivity Simulator
(Indian Meal Presets from IFCT 2017)
24-Hour Interactive Timeline
(Actual vs 120-min Forecasts)
Transparent Driver Panel
(Top clinical contributors)